Chapter 2 • Logarithms

Solved Review Exercise 2

MCQs

1. The standard form of 5.2×10^6 is:
a) 52,000
b) 520,000
c) 5,200,000
d) 52,000,000
Ans: c
Move the decimal six places right.

2. Scientific notation of 0.00034 is:
a) 3.4×10^3
b) 3.4×10^{-4}
c) 3.4×10^4
d) 3.4×10^{-3}
Ans: b
The decimal moves four places right to form 3.4.

3. The base of the common logarithm is:
a) 2
b) 10
c) 5
d) e
Ans: b
Common logarithms use base 10.

4. $\log_2(2^3) =$
a) 1
b) 2
c) 5
d) 3
Ans: d
The logarithm returns the exponent 3.

5. log 100 =
a) 2
b) 3
c) 10
d) 1
Ans: a
$100 = 10^2$.

6. If log 2 = 0.3010, then log 200 is:
a) 1.3010
b) 0.6010
c) 2.3010
d) 2.6010
Ans: c
$200 = 2 \times 100$, so log 200 = 0.3010 + 2.

7. log 0 is:
a) positive
b) negative
c) zero
d) undefined
Ans: d
No real power of a positive base gives zero.

8. log 10,000 =
a) 2
b) 3
c) 4
d) 5
Ans: c
$10{,}000 = 10^4$.

9. log 5 + log 3 =
a) log 0
b) log 2

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c) $\log\frac{5}{3}$
d) log 15
Ans: d
Use the product law.

10. $3^4 = 81$ in logarithmic form is:
a) $\log_3 4 = 81$
b) $\log_4 3 = 81$
c) $\log_3 81 = 4$
d) $\log_4 81 = 3$
Ans: c
Base 3 and exponent 4.

2. Express in scientific notation.
(i) $0.000567 = 5.67\times10^{-4}$.
(ii) $734 = 7.34\times10^{2}$.
(iii) $0.33\times10^{3} = 330$
$= 3.3\times10^{2}$.

3. Express in ordinary notation.
(i) $2.6\times10^{3} = 2600$.
(ii) $8.794\times10^{-4} = 0.0008794$.
(iii) $6\times10^{-6} = 0.000006$.

4. Express in logarithmic form.
(i) $3^7 = 2187 \Leftrightarrow \log_3 2187$
$= 7$.
(ii) $a^b = c \Leftrightarrow \log_a c$
$= b$.
(iii) $12^2 = 144 \Leftrightarrow \log_{12} 144$
$= 2$.

5. Express in exponential form.
(i) $\log_2 8 = x \Leftrightarrow 2^x$
$= 8$.
(ii) $\log_9 729 = 3 \Leftrightarrow 9^3$
$= 729$.
(iii) $\log_4 1024 = 5 \Leftrightarrow 4^5$
$= 1024$.

6. Find x.
(i) $\log_9 x = 0.5$
$\Rightarrow x = 9^{0.5} = 3$.
(ii) $(1/9)^{3x} = 27$
$\Rightarrow 3^{-6x} = 3^3$
$\Rightarrow -6x = 3$
$\Rightarrow x = \frac{-1}{2}$
(iii) $(1/32)^{2x} = 64$
$\Rightarrow 2^{-10x} = 2^6$
$\Rightarrow -10x = 6$
$\Rightarrow x = \frac{-3}{5}$

7. Write as a single logarithm.
(i) $7\log x - 3\log y^2 = \log(x^7/y^6)$.
(ii) $3\log 4 - \log 32 = \log(64/32)$
$= \log 2$.
(iii) $(1/3)\log_5(8\times27) - \log_5 3 = \log_5 6 - \log_5 3$
$= \log_5 2$.

8. Expand the logarithms.
(i) $\log(xyz^6) = \log x + \log y + 6\log z$.
(ii) $\log_3[(m^5n^3)^{\frac{1}{6}}] = (\frac{5}{6})\log_3 m + \frac{1}{2}\log_3 n$.
(iii) $\log\sqrt{(8x^3)} = (1/2)(\log 8 + 3\log x)$
$= (\frac{3}{2})(\log 2 + \log x)$.

9. Evaluate with logarithm tables.

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(i) $N=\sqrt[3]{68.24}$: $\log N=\frac{\log 68.24}{3}\approx\frac{1.8340}{3}=0.6113$
$\Rightarrow N\approx 4.086$.
(ii) $N=319.8\times 3.543$: $\log N\approx 2.5049+0.5494=3.0543$
$\Rightarrow N\approx 1133$.
(iii) $N=\frac{36.12\times 750.9}{113.2\times 9.98}$: $\log N\approx(1.5577+2.8756)-(2.0538+0.9991)=1.3804$
$\Rightarrow N\approx 24.01$.

10. A city population is $p(t) = 22(1.025)^t$ million after 2016. When will it reach 35 million?

$\frac{35}{22} = (1.025)^t$

$\Rightarrow t = \frac{\log(\frac{35}{22})}{\log}(1.025) = 18.8034...$ years.
Nearest whole year: t = 19
calendar year = 2016 + 19.
2035.

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